

Xiangyu Li

PhD in Genetics | Neuroscience & AI Researcher

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Open to Relocation: Shanghai / Hong Kong / Shenzhen / North America *Visa*

Sponsorship Required for U.S./Canada

PROFESSIONAL SUMMARY

Interdisciplinary Ph.D. Researcher with 6+ years of experience bridging **wet-lab neuroscience** (electrophysiology, viral tracing) and **AI-driven behavioral analysis**. Dedicated to leveraging digital biomarkers to accelerate drug discovery for neurodevelopmental disorders.

- **Scientific Impact:** First-author publication in *Molecular Psychiatry* (IF: 10.1)
- **AI & Digital Biomarkers:** Developed markerless pose estimation platforms (DeepLabCut) for high-throughput behavioral phenotyping and optimized biomedical LLM alignment (RLHF) at ByteDance.

EDUCATIONAL BACKGROUND

Central South University | Hunan, China PhD in Genetics | Sep 2019 – Nov 2025

- Advisors: Prof. Jiada Li & Prof. Yousheng Shu
- Focus: Molecular mechanisms of neurodevelopmental disorders (PWS, Autism) and neural circuit modulation.

Central South University | Hunan, China Bachelor of Science in Biological Science | Sep 2015 – June 2019

- **Honors:** Outstanding Graduate; iGEM Silver Medalist (Team Leader, 2018).

PUBLICATIONS

- **Li, X.** ^{#1}, Bader, I. M. I. ^{#1}, Li, X., et al. (2025). Loss of Necdin causes social deficit and aberrant synaptic function through destabilization of SynGAP. *Molecular Psychiatry*. Online ahead of print. **(Q1, IF: 10.1)**
- Liu, D., Xie, Z., Gu, P., **Li, X.**, et al. (2023). Cry1Δ11 mutation induces ADHD-like symptoms through hyperactive dopamine D1 receptor signaling. *JCI Insight*, 8(16): e170434. **(Q1, IF: 6.1)**
- Mu, C. ^{#1}, Liu, P. ^{#1}, Liu, L. ^{#1}, Wang, Y. ^{#1}, Liu, K., **Li, X.**, et al. (2024). KCTD10 p.C124W variant contributes to schizophrenia by attenuating LLPS-mediated synapse formation. *PNAS*, 121(48):e2400464121. **(Q1, IF: 9.4)**
- Gao, Q. ^{#1}, Tian, R. ^{#1}, Han, H., Slone, J., Wang, C., Ke, X., Zhang, T., **Li, X.**, et al. (2022). PINK1-mediated Drp1S616 phosphorylation modulates synaptic development and plasticity via promoting mitochondrial fission. *Signal Transduct Target Ther*, 7: 103. **(Q1, IF: 39.3)**
- **Li, X.**, & Li, J. (2023). "Social Behavior Experimental Methods". *Animal Behavior Experimental Methodology* (Chapter IX). Jul Publisher.

PROFESSIONAL EXPERIENCE (Industry & AI)

ByteDance (Doubao Team) | Remote Bio-AI Specialist (STEM Expert Committee) | July 2025 – Present

- **LLM Alignment for Biomedicine:** Spearheaded the construction of high-quality SFT (Supervised Fine-Tuning) and RLHF (Reinforcement Learning from Human Feedback) datasets specifically for life sciences.
- **Hallucination Mitigation:** Designed 50+ complex prompt chains and "Ground Truth" benchmarks for molecular biology and neuroscience, significantly reducing model hallucinations in long-chain logical reasoning tasks.
- **Domain Evaluation:** Established a rigorous evaluation benchmark for biomedical queries, enhancing the model's credibility in scenarios like gene editing experimental design and medical consultation.

Deer Technology (Chongqing) Co., Ltd. | Remote R&D Lead, Computational Neuroscience | Jan 2024 – Jan 2025

- **Digital Biomarker Platform:** Led the development of a markerless pose estimation system using DeepLabCut (DLC) and PyTorch, enabling high-throughput tracking of multi-animal social interactions.
- **Algorithm Optimization:** Engineered custom features for gait analysis (via foot pressure) and complex behavior recognition (grooming, rearing), successfully filtering noise in dynamic experimental environments.
- **Translational Application:** Collaborated with hardware teams to deploy the system for commercial behavioral phenotyping, ensuring robustness across varying lighting and enclosure conditions.

RESEARCH EXPERIENCE (Academic)

School of Life Sciences, Central South University | Graduate Researcher | Sep 2019 – Present

- **Project:** Molecular & Neural Mechanisms of Social Deficits in Prader-Willi Syndrome.
- **Molecular Mechanism:** Elucidated a novel molecular pathway where **Necdin** regulates the stability of the synaptic protein **SynGAP** via the **SGT1-HSP90 chaperone complex**.
- **Electrophysiology:** Independently established a Whole-cell patch-clamp platform..
- **Circuit Tracing & Manipulation:** Utilized stereotaxic viral injections (AAV and AdV) and optogenetics to map relevant neural circuits.
- **Behavioral Assays:** Designed and executed a battery of behavioral tests (Three-chamber social test, Novel object recognition) to validate phenotypic rescue.

CSU_CHINA iGEM Team Team Leader & Advisor | Jan 2018-Nov 2021

- **Leadership:** Founded the team and secured \$11k+ in funding. Led a 12-member interdisciplinary team to win a **Silver Medal** (2018) in Boston; advised subsequent teams to achieve **Gold Medals** (2019-2020).

Skills

- **Wet Lab: Electrophysiology & Imaging**
 - Electrophysiology:** Whole-cell / Cell-attached patch-clamp (Slices & Cultured neurons).
 - Imaging:** Confocal Microscopy, Super-resolution Imaging (Airyscan), Calcium Imaging basics.
 - Surgery:** Stereotaxic Viral Injection, Cannula Implantation.
 - Molecular:** Western Blot, qPCR, IHC/IF, Mouse Genetics (Cre-LoxP system).
- **Computational Biology & AI:**
 - Languages:** Python (Advanced), R, MATLAB.
 - Frameworks & Tools:** PyTorch, DeepLabCut (Pose Estimation), LLM Fine-tuning (RLHF/SFT principles), ImageJ, Imaris, PyMOL.
 - Data Analysis:** High-content behavioral analysis, Automated tracking, Statistical modeling (SPSS, GraphPad Prism).
- **Core Soft Skills:**
 - Cross-functional Collaboration (AI + Bio), Grant Writing, Project Management, Bilingual Communication (English/Chinese).